

Homework 3: Single Variable Differentiation and Approximation

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Due: Chapter 3, ETC: 1-2 hours, 100 points

Practice Problems: Work together in class

1. Differentiate each of the following functions.

(a) $f(x) = 4x^3 - 6x^2 + 8x - 10$

(b) $g(x) = \frac{12}{x} + 5x^2$

(c) $h(x) = x^{1/2} + x^{-2}$

(d) $k(x) = 7e^x - 3 \ln x$

2. Suppose a firm's total cost function is

$$C(q) = 100 + 20q + 2q^2.$$

(a) Find the marginal cost function $C'(q)$.

(b) Find $C'(10)$.

(c) Interpret $C'(10)$ economically.

(d) Find the second derivative $C''(q)$ and briefly explain what it tells us about marginal cost.

3. Suppose inverse demand is given by

$$p(q) = 80 - 2q.$$

Total revenue is

$$R(q) = p(q)q.$$

(a) Write total revenue $R(q)$ as a function of q only.

(b) Find marginal revenue $R'(q)$.

(c) Find the output level q^* at which marginal revenue is zero.

(d) Use the second derivative to determine whether revenue is maximized or minimized at q^* .

4. Find the critical points of each function and use the second derivative test to determine whether each critical point is a local maximum, local minimum, or whether the test is inconclusive.

(a) $f(x) = x^2 - 6x + 10$

(b) $g(x) = -x^2 + 8x + 3$

(c) $h(x) = x^3 - 3x$

5. Suppose utility from income m is

$$u(m) = \ln m.$$

(a) Find $u'(m)$.

(b) Find $u''(m)$.

(c) Is $u(m)$ increasing?

(d) Is $u(m)$ concave?

(e) Give a brief economic interpretation of $u'(m)$ and $u''(m)$.

Homework Problems: Submit individually

6. Differentiate each of the following functions.

(a) $f(x) = 3x^4 - 2x^3 + x - 9$

(b) $g(x) = \frac{x^2 + 1}{x}$

(c) $h(x) = e^{2x} + 4 \ln x$

(d) $k(x) = (5x^2 + 1)^3$

(e) $m(x) = x^2 e^x$

7. Suppose a firm's production function is

$$Q(L) = 20L^{1/2},$$

where L is labor.

(a) Find the marginal product of labor $Q'(L)$.

(b) Find $Q'(25)$.

(c) Find $Q''(L)$.

(d) Is marginal product increasing or decreasing in L ?

(e) Give an economic interpretation of the sign of $Q''(L)$.

8. Suppose a firm's profit function is

$$\pi(q) = 100q - 4q^2 - 50.$$

(a) Find $\pi'(q)$.

(b) Solve the first-order condition $\pi'(q) = 0$.

(c) Find $\pi''(q)$.

(d) Use the second derivative test to determine whether the critical point is a local maximum or local minimum.

(e) What is the maximum profit?

9. Consider the implicit equation

$$xy = 20.$$

Suppose this equation implicitly defines y as a function of x .

(a) Use implicit differentiation to find $\frac{dy}{dx}$.

(b) Solve explicitly for y as a function of x .

(c) Differentiate the explicit function from part (b).

(d) Verify that the two methods give the same derivative.

(e) In a consumer theory interpretation, suppose $xy = 20$ is an indifference curve. What is the slope of the indifference curve at $(x, y) = (4, 5)$?

10. **Slightly harder problem.** Suppose output is produced according to

$$Y = F(K) = 10K^{1/2}.$$

We want to approximate the change in output when capital increases from $K = 100$ to $K = 104$.

(a) Find $F'(K)$ and $F''(K)$.

(b) Compute the exact change

$$\Delta Y = F(104) - F(100).$$

(c) Use the linear approximation around $K = 100$,

$$F(K) \approx F(100) + F'(100)(K - 100),$$

to approximate $F(104)$.

(d) Use the quadratic approximation around $K = 100$,

$$F(K) \approx F(100) + F'(100)(K - 100) + \frac{1}{2}F''(100)(K - 100)^2,$$

to approximate $F(104)$.

(e) Which approximation is closer to the exact value?